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CS-499: Computer Science Capstone

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Professional Self-Assessment

Based on developing my ePortfolio and coursework experience throughout the Computer Science program at Southern New Hampshire University, I was able to cultivate software organizational and architectural knowledge having worked through a variety of projects related to databases, algorithms & data structures, software security, and object oriented programming principles. I was able to become aware of and learn more about trending technologies that are highly in demand in the industry as of late, such as: artificial intelligence, cloud computing, cybersecurity, data science, etc–which helped me align my career goals and aspirations to what the modern job market is looking for. So far, I am interested in studying cloud computing further, as I would like to dive deeper into strategies on how software services are able to scale from thousands of users to millions that are operating the service at once, making it reliable and efficient.

A specific example in the program outside of the artifacts included in my ePortfolio that helped me develop my IT skills, confidence as well as shape my career aspirations is the UI/UX design and development project, which helped me gain an understanding in the importance of content allocation and button placement of the user interface. The course taught me how to think from the perspective of the user and to consider what essential tools would be worth including in the software and what could be left out in order to prevent using up real estate. It taught me the

importance of applying simplicity and minimalism towards UI/UX design as well as codebase architecture.

Another example that helped me develop my IT skills and software engineering framework was undergoing the software development lifecycle coursework. In this course, I learned the history of how developers formerly implemented the waterfall method in software development and how rigid and inefficient it can be in a rapidly changing technological landscape of today; hence, the agile method is the most commonly adopted method for designing, building, and maintaining software. This particular course highlights the concept that software development is an iterative process and allows for flexibility in the order for when code should be tested, evaluated, and refactored.

In terms of effectively collaborating in a team environment, GitHub works well when it comes to sprint planning and having a backlog of commits to a codebase. Pull requests allow for developers to make the appropriate enhancements to code after a code review, which works seamlessly in a Scrum team environment that is in routine communication with stakeholders that demands a set of criteria to be met based on an established software requirements document.

In the DAD 220 course, I was able to gain a better understanding of relational vs. non-relational databases and how they interact with the frontend as well as the backend. This was done through learning noSQL syntax by using the terminal, applying CRUD operations with basic tables and further learning more about how tables can be joined. In the CS 300 course, I was able to learn extensively about the subject of data structures & algorithms and about how important it is to consider time & space complexity when handling data. If handled incorrectly, programs will run very slow, and can crash in some cases. Therefore, utilizing the correct data structure and algorithm–especially when more users are active on the software application or when high

latency is required–is an important aspect to consider. In the CS 405 course, I learned about secure coding techniques and best practices in order to prevent potential SQL injection on code that lacks the appropriate input checks. Not only that, but I have learned about the Defense in Depth concept, which emphasizes the need for multiple layers of security which include the perimeter, network, endpoint, application, and data.

The artifact that I decided to use for my ePortfolio addresses each of the three categories for this course in one Android Event Planning mobile application that was initially created in the CS 360: Mobile Architect & Programming course. The Event Planning application was designed for users to be able to store events along with the title, date, description, and phone number of the event. Users are able to create, read, update, and delete any of the events as they see fit from the frontend UI of the application, which makes the appropriate interactions with the SQLite database within the backend. Once an event is stored and happens to be within 7 days of the current day, the application will send a text notification to the phone number that was saved along with the event. This function acts as a reminder for recipients that are interested in the event. The application also implements a user login system that stores a username and password within the SQLite database.

Having completed all the enhancements in accordance with each of the three categories in key areas such as: software engineering & design, data structures & algorithms, and databases, I was able to utilize all the skills and knowledge that I have gained in the computer science program by implementing input validation techniques, applying an MVC architecture, write industry standard comments clearly explaining the intent of each function/method, min heap (priority

queue) data structure for event notification, and the PBKDF2 (Password-Based Key Derivation Function 2) for password hashing.